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Climate change impact assessment on livelihood vulnerability of tribal communities: a systematic review

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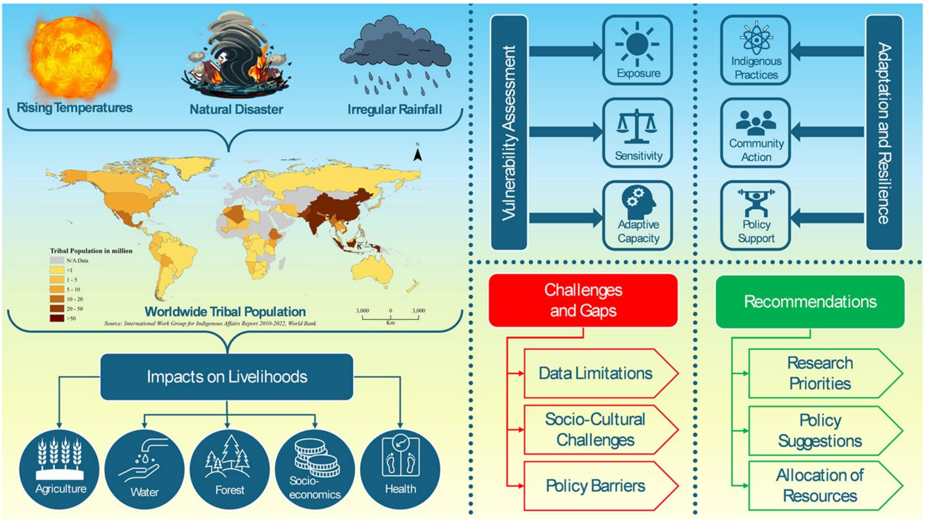
Abstract
Climate change and variability have significantly affected livelihoods across different geographical areas over various time scales; however, major risks have been faced by tribal communities and marginalised populations. Despite the significant impacts of climate change on tribal communities, there is a notable gap in multidisciplinary analyses that integrate indigenous knowledge with scientific approaches. Therefore, it is necessary to understand the complex relationship between climate change and the vulnerability of tribal communities. The review evaluates the impact of climate change on tribal livelihoods, identifying key vulnerability indicators and proposing effective adaptation and resilience strategies. Tribal communities, constituting 5% of the world's population, face increased vulnerability due to climate change, influenced by factors such as education, income levels, natural resource dependence, and governance effectiveness. Worldwide case studies illustrated the different impacts of climate change on tribal communities and their adaptive strategies. The literature revealed that Indigenous knowledge and practices, community-based adaptation approaches, and ecosystem-based methods are essential in building resilience. However, data limitations, policy and implementation barriers, and socio-cultural challenges persist, impeding effective adaptation. Integrating Indigenous knowledge with scientific methods, enhancing data collection methodologies, and fostering inclusive policy frameworks are essential for improving tribal resilience. This study recommends policies for Indigenous rights recognition, community-driven projects' support, and capacity-building programs' prioritisation. It is also suggested that stakeholders, researchers, policymakers, and community members collaborate to address the multifaceted challenges of climate change and ensure sustainable futures for tribal communities.

Extended author information available on the last page of the article

Graphical abstract



Keywords Climate change · Tribal communities · Livelihood vulnerability · Mitigation and adaptation strategies · Indigenous knowledge

1 Introduction

Climate change (CC) describes long-term shifts in temperature, precipitation patterns, and other atmospheric variables on Earth (Kuniyal et al. 2021; Singh et al. 2024). This occurrence has been mainly caused by human activity, specifically releasing of greenhouse gases (GHGs) such as methane (CH_4), carbon dioxide (CO_2), and nitrous oxide (N_2O) (Li et al. 2024). In comparison, climate variability discusses short-term variations in climate constraints around the long-term mean, including phenomena like El Niño and La Niña, which cause significant fluctuations in weather patterns over seasons and years (Lee et al. 2022; Singh et al. 2023a). While CC agrees with the warming trend, variability proposes complexity and unpredictability in weather events (Scafetta 2024; Singh et al. 2023b). The Intergovernmental Panel on Climate Change (IPCC) has projected that the global temperature will increase by 1.5 °C by the end of the 21 st century, which will have profound consequences for natural and human systems (Kumar and Mohanasundari 2025b; Phuong et al. 2023). These extensive changes are affecting environments and communities globally.

CC and variability have significantly affected livelihoods across different geographical areas over various time scales. The United States (U.S.) has faced more frequent extreme hot temperatures and precipitation events, with a fall in extreme cold events, especially in northern regions (Monier and Gao 2015). Rural communities are more vulnerable than urban areas due to the demographics, literacy, occupations, incomes, prevalence of poverty, and reliance on government funding (Demissie and Kasie 2017). However, CC effects may vary with region and economic sector, with some rural areas benefiting while others suffer. Climatic conditions in the Northeast U.S. might be advantageous for agricultural and forestry activities, whereas the Southwest and Southeast could face heightened water stress and energy costs, respectively (Lal et al. 2011). It is projected that Latin America, and the Caribbean will experience a mean temperature increase of up to 4.5 °C by the end of the 21 st century, compared to pre-industrial (Reyer et al. 2017) due to changed rainfall patterns, heat extremes, drought, and rising sea levels. These changes will influence agriculture, livestock, fisheries, and water resources, although some opportunities, like increased rice yields and fish catch capacity, may rise in specific regions. CC is expected to increase global temperatures past + 1.5 °C in the coming decades, potentially boosting summer tourism and reducing electricity demand in Western Europe (Jacob et al. 2018). However, the frequency and intensity of extreme events increased due to CC, directly impacting human health through injuries and fatalities and indirectly through effects on water, food, air quality, mental well-being, productivity, and harm to residential and health services in Australia (Lansbury Hall and Crosby 2022). Sub-Saharan Africa suffered undernutrition, disease outbreaks, and agricultural vulnerabilities, leading to urban migration and increased food prices due to CC (Serdeczny et al. 2017). Similarly, South Asia faced a lack of freshwater supply, glaciers melting, reduced agriculture productivity, loss of biodiversity, poor human health, extreme weather events, and resource strain due to climate-induced immigration from neighbouring countries (Verma 2021). Furthermore, increasing mean annual temperatures decrease land productivity for most crops, jeopardising the food safety of India's small and marginal farming households (Praveen and Sharma 2020).

CC is a significant global issue that disproportionately impacts vulnerable communities, especially those reliant on natural resources for livelihood (Sahoo et al. 2023). Tribal communities, often living in environmentally sensitive and marginal regions, are most affected (Kumar et al. 2023). The community heavily depend on agriculture, hunting, fishing, gathering, forestry, and other natural resource-based activities, which makes them highly vulnerable to the detrimental impacts of CC and variability. CC events have been reported to substantially impact the ecosystems crucial for tribal livelihoods (Pratap and Pratap 2021). As a result, tribal communities face threats to food security, water availability, health, and socioeconomic stability due to CC and retain traditional knowledge and practices for resilience construction (Mallick et al. 2024; Rahmah and Sulistyono 2024). Despite their importance, mainstream climate adaptation strategies often overlook these practices.

CC studies consistently explore the concept of vulnerability as a fundamental and evolving concept. Vulnerability to CC-induced natural disasters is divided into social, physical, economic, and environmental categories (Kasthala et al. 2024). The IPCC's 4 th Assessment Report specifies vulnerability as "the degree to which a system is susceptible to and unable to cope with adverse effects of climate change, including climate variability and extremes (IPCC 2007). Vulnerability is also a function of the character, magnitude, and rate of climate variation to which a system is exposed, its sensitivity, and its adaptive capacity" (IPCC 2007; Kasthala et al. 2024). Figure 1 shows the key dimensions influencing the vulnerability of tribal communities to CC. Tribal communities face increased exposure to climate variability due to geographical isolation and inadequate infrastructure, significant sensitivity due to high dependence on climate-sensitive resources, and inadequate adaptive capacity due to socioeconomic constraints, highlighting their exposure to climate impacts.

The literature on CC vulnerability, while extensive, lacks a focus on tribal livelihood vulnerability assessment in the context of changing climatic conditions (Berrang-Ford et al. 2015; Crane et al. 2017; Kasthala et al. 2024; Kumari et al. 2023; Räsänen et al., 2016;

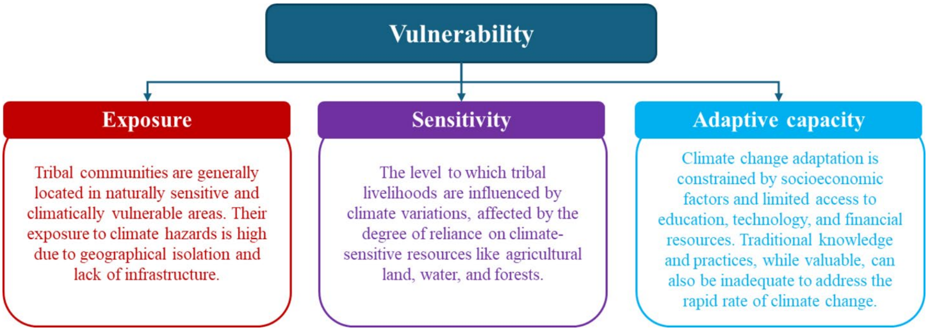


Fig. 1 Contributing factors of livelihood vulnerability

Singh et al. 2017). However, many researchers/scholars still use the Department of International Development's (DFID) livelihood framework. Expanding on this model, Hahn et al. (2009) developed the Livelihood Vulnerability Index (LVI), a promising indicator-based technique for assessing livelihood vulnerability, incorporating elements from DFID and IPCC frameworks. Furthermore, Reed et al. (2013) incorporated theoretical understanding from analyses of sustainable livelihoods and analytical frameworks to investigate the livelihood vulnerability to CC. Researchers have also explored various methodologies to evaluate livelihood vulnerability; however, the literature lacks a universal framework for all ecosystems and geographies (Kasthala et al. 2024; Kumari et al. 2023). With this research gap, it is necessary to understand/develop a comprehensive framework that can correctly evaluate the vulnerability of tribal livelihoods across various ecosystems and geographies. This framework should combine climatic, socio-economic, and ecological variables and be acceptable to local contexts. Understanding these nuanced vulnerabilities is essential for designing effective adaptation strategies and policies to mitigate CC's adverse impacts on tribal communities. Therefore, this study investigates the livelihood vulnerability of tribal communities in the context of CC, providing a critical tool for policymakers and practitioners to enhance resilience among these vulnerable populations. The primary objectives of this study are to (i) examine the impacts of CC and variability on the livelihoods of tribal communities, (ii) assess the vulnerability of tribal communities to CC, (iii) explore adaptation and resilience strategies, and (iv) provide future directions and policy recommendations. A comprehensive conceptual framework for the present study is provided in Fig. 2.

2 Methodology

The literature search strategy followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) model (Albhirat et al. 2024),which guides the recognition of relevant studies. The process involved systematic searches of electronic databases, comprising Google Scholar, Web of Science, Scopus, and PubMed, and institutional repositories (Fig. 3), including literature from governmental and non-governmental organisations and reports from international bodies (e.g., UN, IPCC). The search terms were developed based on the research objectives. They integrated combinations of keywords related to "climate change", "climate variability", "tribal communities", "Indigenous populations", "tribal live-

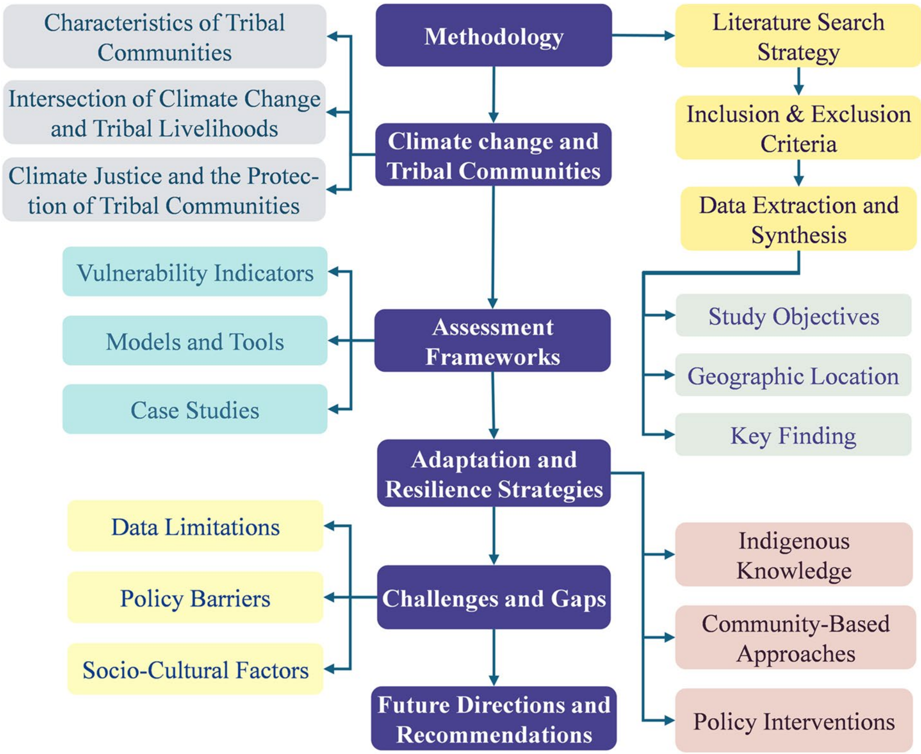


Fig. 2 Conceptual Framework of Livelihood Vulnerability in Tribal Communities

lihood", "livelihood vulnerability", "adaptation strategies", "Indigenous knowledge", and synonyms or variations thereof. Boolean operatives such as "AND" and "OR" were used to enhance search queries.

Studies considered for inclusion were peer-reviewed articles, review papers, case studies, reports, and academic papers published in English for 20 years (from 2005 to 2024). Older seminal works were also considered to prepare foundational understanding where required. Although non-English publications were excluded, this may have resulted in the exclusion of essential studies, especially those conducted by experts in the area who had better access to tribal communities. This decision was made because of practical limitations, including language proficiency limitations and the absence of credible translation services, which could affect consistency and accuracy. The review focused on English-language materials to maintain rigorous methodology and assure quality control. It has acknowledged this limitation and recommends that future analyses include non-English literature by collaborating with multilingual scholars or using translation assistance. Furthermore, grey literature, articles lacking empirical data or robust theoretical frameworks, and studies not directly or indirectly related to the vulnerability of tribal livelihoods due to CC were excluded from the review. After exclusion, out of the initial 267 articles/reports identified, 169 articles/reports met the predefined criteria related to tribal livelihood vulnerability due to CC and were included in the review. Additionally, over 97 articles/reports were reviewed to provide a comprehensive background on the global CC scenario and its associated issues. Data

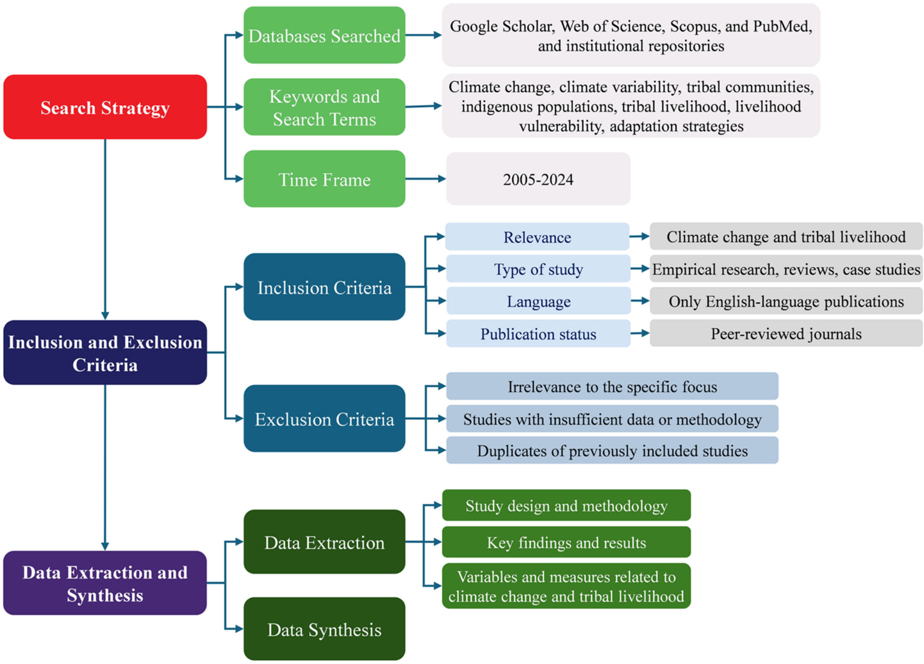


Fig. 3 Methodology Framework

extraction and synthesis were conducted following the PRISMA model guidelines. Using a standardised data extraction form, the study systematically extracted and comprehensively arranged pertinent data from chosen studies, such as study features, goals, main inferences, and methodological specifics. The thematic analysis identified common themes, patterns, and modifications across the literature. Results were synthesised narratively and enhanced by tabular summaries and visual representations where related.

3 Climate change and tribal communities

3.1 Characteristics of tribal communities

Tribal communities, also known as indigenous peoples, constitute about 5% of the world's population (Das and Basu 2022) (Fig. 4). The basic characteristics of tribal communities are shown in Fig. 5. Several environmental challenges, such as deforestation, land degradation, and biodiversity loss, have been faced by tribal communities (Kumar et al. 2022). The challenges faced by tribal groups have been intensified by colonial and post-colonial policies that often marginalized them and limited their access to traditional lands and resources (Chiweshe 2023). The CC impacts add another layer of difficulty, threatening the sustainability of their traditional livelihoods (Parrotta and Agnoletti 2012). There is great ethnic diversity that appears in the broad range of economic opportunities among tribal groups in India. According to the 2011 Census, tribes comprise 8.6% of India's population (Das and Basu 2022). In many developing countries such as India, Indonesia, Nepal, Brazil, and Africa,

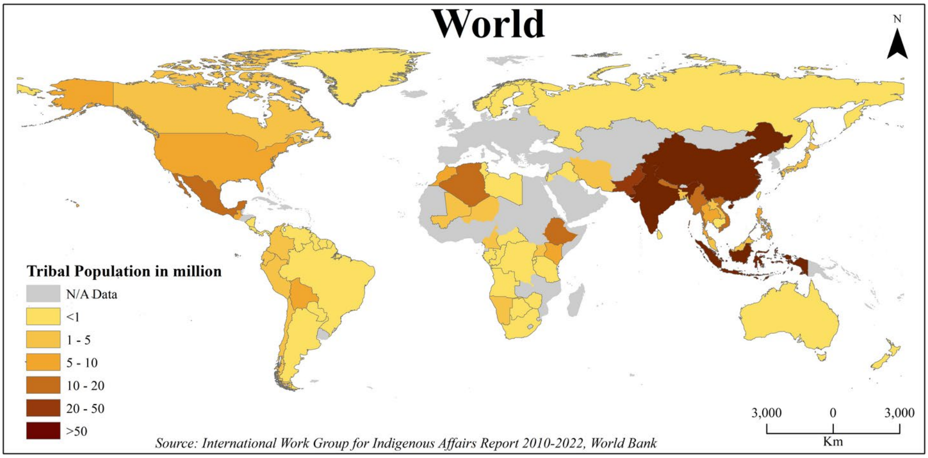


Fig. 4 Worldwide distribution of the Indigenous population

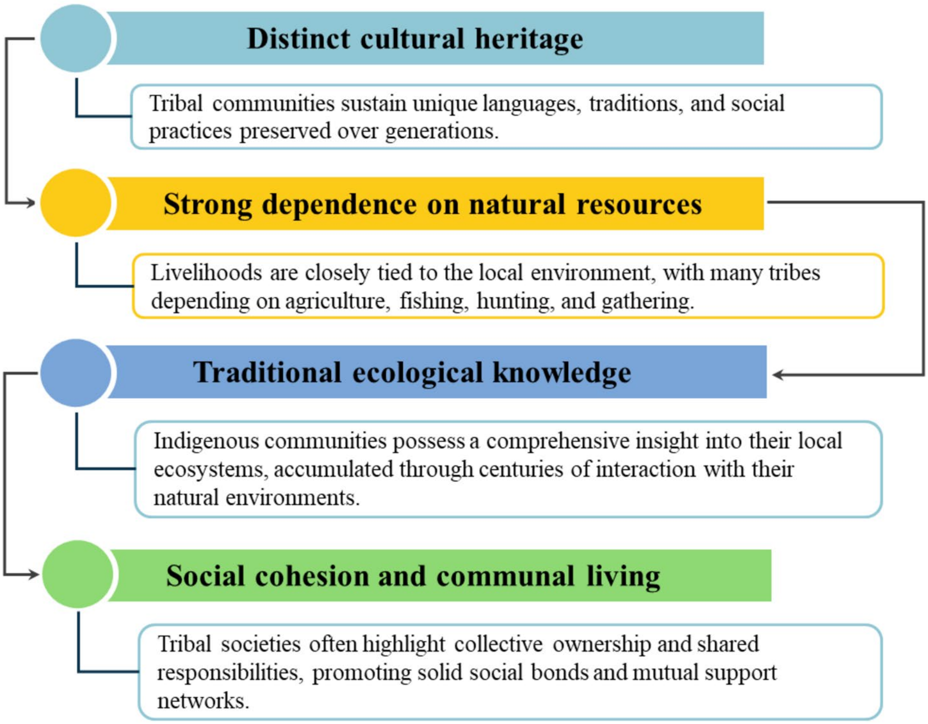


Fig. 5 Common characteristics of tribal communities

tribal peoples depend primarily on forest products such as food, fuel wood, fodder, lumber, and medicinal plants (Kumar and Saikia 2020). These resources are not only for generating financial income but also for meeting their subsistence needs. Tribal communities, despite their rich cultural heritage, frequently face socioeconomic marginalization, limited access to education and healthcare, and political exclusion (Panda et al. 2022), which increase their susceptibility to CC.

3.2 Intersection of climate change and tribal livelihoods

The relationship between CC impacts and the livelihoods of tribal communities is complex and dynamic (Mallick et al. 2024). Tribal communities are among the most vulnerable to CC due to their heavy dependence on natural resources, marginal socioeconomic status, and often limited adaptive capacity (Hazarika et al. 2024; Kumar and Mohanasundari 2025a). The complex impacts of CC on tribal livelihoods include agricultural practices, water resources, forest and biodiversity dependence, health and wellbeing, and socioeconomic implications, as discussed below.

3.2.1 Agricultural practices

Agriculture is the backbone of tribal economies, ensuring food security and livelihoods for millions (Nautiyal and Goswami 2022). Tribal communities are involved in cultivating crops and raising livestock to meet their basic needs (Ramakrishnan et al. 2024). However, changing climatic patterns have disturbed traditional agricultural practices, reducing crop yields (Aich et al. 2022). Several studies have reported the loss of traditional crops, decreased crop yields, and increased pest and disease pressures in many tribal regions (Aich et al. 2022; Atta et al. 2023; Grigorieva et al. 2023; Hussain et al. 2020; Kar et al. 2024; Lin et al. 2008; Praveena and Malaisamy 2024). Tropical Asia suffered crop losses and decreased rice and wheat productivity in tropical Asia due to changing monsoon patterns, which influenced the timing and distribution of rainfall (Huggi et al. 2024). The reduced rainfall in American monsoon regions, including southern Chile, southwestern Argentina, southern Peru, and west-central America, has negatively impacted the productivity of major crops like wheat, soybean, and maize. In addition, maize productivity has declined from 0.01 to 0.05 tonnes yearly in the U.S., primarily attributed to internal variability in the El Niño-Southern Oscillation (ENSO). The Sahel region experienced a significant decrease in groundnut productivity from 850 kg/ha in 1966-67 to 440 kg/ha in 1981, mainly as a result of variations in the distribution of rainfall during growing seasons (Huggi et al. 2024). Studies also reported that grasslands in Africa are at risk due to monsoon variability, reducing fodder availability for livestock, and rising temperatures accelerating pest and disease spread, causing further crop failures (Grigorieva et al. 2023). Extreme weather events like floods and droughts can lead to soil erosion and nutrient depletion, reducing agricultural productivity (Kulkarni 2021).

3.2.2 Water resources

Tribal communities reliant on rivers, lakes, and groundwater are at a higher threat due to CC, which disrupts hydrological cycles and directly threatens their traditional water sources, undermining not only daily household needs but also agricultural and livestock activities critical to their livelihoods (Soomro et al. 2024). Reduced water availability during dry seasons (like droughts) and increased frequency of extreme weather events (for example, floods) pose significant sustainability challenges (Shiva Shankar et al. 2021). Singh and Goyal (2025) and Tzanakakis et al. (2020) revealed that water scarcity is a pressing global issue. More than 2 billion people have inadequate access to safe drinking water and reside in locations with severe water stress. 3.4 million die annually from contaminated water and 50% of hospital beds are filled with affected individuals. Unlike urban populations or mainstream rural areas, many tribal regions are geographically isolated, lacking investments in modern water management technologies, piped systems, and disaster preparedness mechanisms. Therefore, even developments in infrastructure and innovations enhancing water use efficiency avoid tribal areas, increasing disparities. Population growth, improved lifestyles, and rising temperatures have increased household water usage, causing water quality issues for drinking, agriculture, and other purposes (Cozzetto et al. 2013). However, for tribal communities, these pressures are intensified by socio-political marginalization, limited policy attention, and dependence on climate-sensitive ecosystems, making them disproportionately affected by deteriorating water resources.

3.2.3 Forest and biodiversity dependence

Many tribal communities depend on forests and biodiversity for their livelihoods, including food, fuel, medicine, and cultural practices, but CC threaten their resilience, impacting forest dynamics, species, and ecosystem services (Azhar et al. 2022; Voggesser et al. 2013). For tribal populations, this changes to the depletion of NTFPs such as wild fruits, medicinal plants, honey, and fuelwood, essential for their subsistence economy and cultural identity. CC has affected the timing of flowering, fruiting, and seed dispersal, disrupting ecological connections and traditional harvesting customs (Hamilton et al. 2024). Moreover, amplified frequency and intensity of deforestation, forest fires, pests, and diseases have further destroyed forest ecosystems. These disturbances reduce the availability of forest resources, limit access to traditional medicines and food sources, and force tribal populations to travel longer distances or adopt alternative livelihoods, often outside their cultural contexts (Kumar et al. 2022). Approximately 20–25% of global greenhouse gas emissions are attributable to deforestation; one million species are in danger of extinction (Pal 2022). While these figures show global concerns, their immediate impact is disproportionately felt by forest-dependent tribal communities whose socio-economic and cultural well-being is closely tied to the health of forest ecosystems.

3.2.4 Health and wellbeing

CC leads to extreme weather events, increasing vector-borne diseases, waterborne illnesses, malnutrition, and mental health issues among tribal populations (Chandra and Mukherjee 2022; Donatuto et al. 2020; Kumar and Mohanasundari 2025c). Rising temperatures raise the risk of heatstroke and dehydration, particularly for vulnerable populations like older adults and children (Ndlovu and Chungag 2024). For tribal communities, limited access to modern healthcare infrastructure and dependence on traditional medicine makes them more vulnerable to these health risks. Prolonged droughts can lead to water scarcity and food insecurity, strengthening the risk of malnutrition and related health issues (Jones 2019; Ngcamu and Chari 2020). Many tribal communities already face higher rates of undernutrition due to socio-economic marginalization, and climate-induced crop failures further worsen these conditions. Similarly, floods and cyclones can result in water contamination, displacement, and injuries, aggravating health vulnerabilities among tribal communities (Suar et al. 2024). The lack of social safety nets and post-disaster recovery processes in tribal areas increases these health issues, making them more evident compared to other population groups.

3.2.5 Socioeconomic implications

CC impacts tribal livelihoods, affecting household incomes, employment opportunities, social cohesion, agricultural productivity, natural resource-based livelihoods, and traditional knowledge systems, causing financial instability and resilience (Bora and Mahanta 2024; Rautela and Karki 2015). Unlike mainstream populations, tribal communities' socio-economic systems are directly connected to natural resources and subsistence economies, making them more vulnerable to climate hazards. Furthermore, climate-induced displacement and migration can drain social networks and augment conflicts over resources and land (Warner and Wiegel 2021). The breaking down of social networks is especially problematic for tribal people as these networks usually function as informal safety nets in times of crisis. Tribal communities' vulnerability to CC is aggravated by limited access to markets, financial resources, and social services (George et al. 2023). Traditional practices are at risk of cultural loss as younger generations may abandon them due to their unsustainable nature (Abdullah and Khan 2023).

3.3 Climate justice and the protection of tribal communities

The concept of climate justice demonstrates the moral duty to preserve people who are least responsible for global greenhouse gas emissions yet bear the most severe effects (Okereke 2010). Tribal populations frequently contribute less to the drivers of climate change. Still, they are at the forefront of its consequences due to poor adaptive capacity, limited access to institutional support, and marginalization from mainstream policy frameworks (Norton-Smith 2016). Investment in the protection and resilience of tribal communities is an issue of social equity and an essential component of achieving climate justice (Bouyé et al. 2021). Furthermore, their deep repositories of traditional knowledge systems, sustainable resource management approaches, and community-driven adaptive strategies offer important lessons for broader climate adaptation efforts (Mallick et al. 2024). Integrating indigenous knowledge with current scientific techniques produces more comprehensive and culturally suitable adaptation frameworks (Taylor et al. 2023). The study indicates the importance of protecting minority tribal communities by addressing their vulnerabilities and their ability to contribute to global efforts to build climate resilience. Ensuring the participation of tribal voices in policy and adaptation initiatives connects with the broader aims of justice, equity, and sustainability in climate governance (Carmona 2024).

4 Vulnerability assessment frameworks

Vulnerability assessment frameworks (VAFs) are essential to understanding the relationships between CC impacts and tribal livelihoods. Frameworks assess the vulnerability of tribal communities to climate-induced disasters, providing insights into contributing factors and recommending targeted adaptation and resilience strategies. Therefore, it is necessary to discuss the key components of VAFs, commonly used indicators, assessment models/tools, and present case studies illustrating their application in worldwide tribal communities.

4.1 Vulnerability indicators

Indicators play a significant role in assessing vulnerability (Kumar et al. 2024; Yadava and Sinha 2020), incorporating multiple dimensions, such as socioeconomic, environmental, and institutional factors collectively affecting communities' adaptive capacity. Socioeconomic indicators are income levels, education, access to healthcare, and social networks, especially the available community resources for coping with CC impacts (Hahn et al. 2009). Environmental factors focus on natural resource dependence, land tenure systems, and ecological resilience, emphasising the vulnerability of tribal livelihoods to changes in climatic conditions (Bora and Mahanta 2024; Long and Steel 2020; Mallick et al. 2024; Ramakrishnan et al. 2024). Institutional factors are the effectiveness of governance policies, structures, and institutional support systems in facilitating adaptation and resilience-building attempts within tribal communities (Mehta 2024). Table 1 presents vulnerability indicators in tribal communities, providing a comprehensive understanding of CC and variability and aiding in developing targeted interventions and adaptation strategies.

4.2 Assessment models and tools

Various assessment models and tools have been developed to measure vulnerability and guide decision-making activities in tribal areas, as presented in Table 2. These models range from qualitative to quantitative approaches. Qualitative methods such as participatory vulnerability assessments and community-based risk mapping prioritise community engagement and local knowledge, allowing tribal communities to identify and select vulnerability factors based on their lived experiences and observations (Smit and Wandel 2006). Quantitative models, such as vulnerability index construction, situation-based modelling, and indicator-based models, use statistical procedures and geographic information system (GIS) mapping to analyse vulnerability hotspots, quantify risk levels, and prioritise intrusion areas (Panda 2017). Integrating quantitative and qualitative methods in indicator-based models, utilizing data from censuses, surveys, and climatic records, provides a comprehensive understanding of vulnerability dynamics (Hahn et al. 2009; Phuong et al. 2023).

Vulnerability Index Tools are often used to quantify vulnerability by aggregating multiple indicators into a single index score. Tools like the Socioeconomic Vulnerability Index (George et al. 2023; George and Sharma 2022) or the Climate Vulnerability Index are commonly used (Ghosh and Ghosal 2021). Participatory assessment methods involve directly engaging communities to recognise their perceptions of vulnerability, adaptive capacities, and specific challenges (Andrachuk and Smit 2012). Participatory assessment tools such as community mapping, seasonal calendars, and vulnerability and capacity assessments are

Table 1 Indicators of livelihood vulnerability in tribal communities

Sl No.	Category	Indicators	Explanation	References
1.	Socio-economic Indicators	Income	• Low income levels frequently suggest limited resources for adaptation and recovery.	Yadava and Sinha (2020)
		Education	• Higher levels of education increase adaptive capacity through improved awareness and access to information.	Das and Basu (2022); Phuong et al. (2023)
		Access to Services	• The availability of healthcare, infrastructure, and social services influences resilience.	Ghosh and Ghosal (2021); Rehman et al. (2022)
		Employment Patterns	• Reliance on climate-sensitive areas such as agriculture or forestry can increase vulnerability.	Das and Basu (2022)
2.	Environmental Indicators	Dependence on Natural Resources	• Dependency on agriculture, forests, water bodies, and biodiversity for livelihoods	Pandey et al. (2017)
		Climate Sensitivity	• Exposure to climate extremes like floods, droughts, or cyclones	Mehzabin and Mondal (2021)
		Geographic Location	• Proximity to disaster-prone areas such as coastal regions, forests, or arid lands	Mehzabin and Mondal (2021)
3.	Cultural and Institutional Indicators	Cultural Identity	• Conservation of traditional knowledge and practices for adaptation.	Makondo and Thomas (2018)
		Social Cohesion	• Strength of community networks and support systems.	Townshend et al. (2015)
		Access to Decision-making Practices	• Inclusion in local governance and policy formulation	Brugnach et al. (2017)
		Legal Rights	• Identification of land tenure and resource ownership rights	Whyte (2013)
4.	Health and Wellbeing Indicators	Health Infrastructure	• Accessibility of healthcare facilities and services.	Das and Basu (2022); Hahn et al. (2009); Venus et al. (2022)
		Prevalence of Diseases	• Vulnerability to climate-related health risks like water and vector-borne diseases or malnutrition	Rai et al. (2022)
		Food Security	• Accessibility and availability of nutritious food sources	Phuong et al. (2023)
		Mortality and Life Expectancy	• Includes child and infant mortality rates, and life expectancy, reflecting overall community health status.	Jaakkola et al. (2018)
		Vaccination Coverage	• Access to immunization programs, reducing vulnerability to preventable diseases.	Asaaga et al. (2021); Johnson et al. (2022)
5.	Adaptive Capacity Indicators	Infrastructure Development	• Access to technologies for climate-resilient agriculture, water management, etc.	Srivastav et al. (2021)
		Capacity Building	• Participating in training, knowledge exchange, and skill development programs can increase adaptation.	Phuong et al. (2023); Venus et al. (2022)
		Financial Resources	• Access to loans, insurance, and social safety nets for recovery post-disaster.	Rai et al. (2022); Venus et al. (2022)

Table 1 (continued)

Sl No.	Category	Indicators	Explanation	References
6.	Governance and Policy Indicators	Policy Support	• Adaptive policies and programs exist at local, national, and international levels.	Reed et al. (2006)
		Institutional Capacity	• Acceptability of institutions for disaster management, climate adaptation, and disaster risk reduction	Birkmann and von Teichman (2010)
		Participation and Inclusivity	• Participation of tribal communities in decision-making practices and policy implementation	Menon and Hartz-Karp (2019)

samples of such attempts (Ghosh and Ghosal 2020; Jha and Negi 2021). GIS are extensively applied to map vulnerability spatially by overlaying climate, environmental, and socioeconomic data (Sarun et al. 2018). GIS tools can assist in finding areas most vulnerable to CC impacts and select intrusions accordingly. Livelihood assessment tools aim to understand the diverse livelihood strategies employed by communities and the impact of CC on these strategies (Ahmad et al. 2023; Hahn et al. 2009; Joshi and Rawat 2021; Venus et al. 2022). Livelihood frameworks assess community vulnerability by examining assets, capabilities, and strategies, while adaptation planning tools identify intervention areas based on vulnerability assessments (Tuler et al. 2020). Cost-benefit analysis and scenario planning are frequently employed to estimate the effectiveness of various adaptation alternatives (Kwakkel 2020).

4.3 Worldwide case studies and applications

CC impacts on Indigenous peoples and local communities are diverse, widespread, and noticeable, emphasising the need for context-specific adaptation strategies and recognising their experiences in addressing loss and damage across different climate zones globally (Reyes-García et al. 2024). Case studies can highlight specific regions, as shown in Table 3. Tribal communities in the coastal areas of the U.S. use effective strategies rooted in cultural practices and values to address climate-induced displacement, highlighting the importance of community-led and government-supported relocation programs (Maldonado et al. 2013). CC presents severe threats to the accessibility and quality of traditional foods that are essential to the survival of American Indian and Alaska Native tribes, influencing their cultural, economic, and social welfare, with varied consequences observed in different ecosystems. (Lynn et al. 2013). In coastal Louisiana, tribal communities facing forced displacement due to environmental changes such as rising sea levels, historical discriminatory processes, and industrial growth need adaptive governance structures that support both in-situ adaptation and community-led relocation, integrating diverse knowledge systems (Maldonado 2014). The Pyramid Lake Paiute Tribe in Nevada illustrates the impact of CC on cultural and economic prosperity by reducing crucial water inflows for the survival of endangered cui-ui fish. The tribe's adaptation to changes is influenced by their commitment to sustainability, resource management, control over invasive species, collaboration with scientists, and awareness of CC (Gautam et al. 2013).

The Saami, who are the only tribal community in the European Union, exhibit similar or even better health conditions in comparison to the neighbouring communities. Still, the

Table 2 Assessment tools and models for evaluating vulnerability

Sl. No.	Assessment Framework/Model	Description	Key Features	References
1.	IPCC Vulnerability Assessment Framework	Developed by the Intergovernmental Panel on Climate Change (IPCC), this framework provides a systematic approach to assess vulnerability to climate change.	Considers exposure, sensitivity, and adaptive capacity	IPCC (2014)
2.	Livelihood Vulnerability Index	A composite index is used to measure livelihoods' vulnerability to climate change.	Incorporates socioeconomic, environmental, and institutional factors	Ahmad et al. (2023); Hahn et al. (2009); Joshi and Rawat (2021); Venus et al. (2022)
3.	Community-Based Vulnerability Assessment	Focuses on participatory approaches involving local communities in assessing their vulnerability to climate change impacts	Emphasises local knowledge and perceptions	Andrachuk and Smit (2012)
4.	Composite Social Vulnerability Index	Measures socioeconomic vulnerability to climate change by integrating social, economic, and demographic variables.	Calculates Infrastructural, Social, and Climate Vulnerability Index	George et al. (2023); George and Sharma (2022)
5.	Socio-ecological vulnerability	Socio-ecological vulnerability assessment to climate change by integrating socio-economic, agriculture, water, and forest	Quantifies exposure, sensitivity, and adaptive capacity	Jha and Negi (2021)
6.	Composite Vulnerability Index	Assessed the local dimensions of vulnerability using a composite vulnerability index based on environmental and socioeconomic factors	Evaluate and map the environmental, socioeconomic, and composite vulnerability index.	Sarun et al. (2018)
7.	Household Vulnerability Index	Evaluate household-level vulnerability to climate change	Includes income loss, crop loss, housing loss, and overall livelihood loss	Ghosh and Ghosal (2020)
8.	Climate Change Vulnerability Index	Examine climate change vulnerability among the agriculture-dependent and forest resource-dependent villages.	Quantifies exposure, sensitivity, adaptive capacity, and climate change vulnerability index	Ghosh and Ghosal (2021)

impact of CC is already evident in their reindeer culture and mental health, making them vulnerable to the changing conditions around them, particularly in the Sápmi region (Jaakkola et al. 2018). However, Climate-induced planned relocations in Latin America and the Caribbean, experienced by the Guna people in Panama and tribal communities in Chiapas, often worsen cultural and socioeconomic problems due to inadequate planning and execution rather than effectively dealing with the challenges of CC (Felipe Pérez and Tomaselli 2021). Mapuche communities in the Araucanía Region of Chile are more vulnerable to CC due to factors such as water scarcity, decreased agricultural productivity, ecological pressure, and the loss of traditional knowledge, imposing local resilience and tailored

Table 3 Case studies on tribal livelihood vulnerability and adaptation

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
1.	Worldwide (48 sites across all climate zones)	Global level	Identify climate change indicators and impacts experienced locally by Indigenous Peoples and local communities.	• Ongoing, tangible, and widespread impacts • Affected multiple elements of social-ecological systems across different climate zones and livelihood activities • Identifying economic and non-economic loss and damage due to climate change through local reports.	Reyes-García et al. (2024)	• Climate change impacts vary according to climate zone and livelihood activity, demonstrating the need for localised adaptation strategies.
2.	United States	National and regional level	Examine the impact of climate change-induced displacement on tribal communities and explore their adaptation strategies.	• Using innovative strategies to mitigate the impacts of forced relocation due to sea level rise, land erosion, and permafrost thaw. • A significant risk of cultural and economic loss, health impacts, and exacerbated poverty and injustice	Maldonado et al. (2013)	• Inadequate governance mechanisms and budgets are major barriers. • Community-led efforts that align with cultural practices are more effective.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
3.	United States	National level	Investigate the impacts of climate change on tribal traditional foods and the cultural, economic, and health implications.	• Significantly affecting native fungi, plant, and animal species essential for tribal traditional foods. • Legal and regulatory relationships with the federal government influence access to these resources.	Lynn et al. (2013)	• Tribal traditional foods are vital to culture and economy • Adaptation strategies involving tribal participation can enhance resilience and inform broader governmental strategies
4.	Alberta, Canada	Local level	Assess the tribal vulnerabilities and adaptive capacity related to climate change.	• Reduces water supplies, impacting Pyramid Lake and the endangered cui-ui fish, vital for cultural and economic sustenance. • Limited economic opportunities and reduced federal support constrain adaptive capacity.	Gautam et al. (2013)	• The tribe shows sustainability-based values, strong technical capacity in natural resource management, proactive invasive species control, robust scientific networks, and high awareness of climate change.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
5.	Louisiana, United States	Regional level	Evaluate the impact of environmental changes, forced displacement, and inadequate governance mechanisms on tribal communities.	• Forcibly displacing tribal communities due to Environmental changes, leading to severe social, cultural, health, and economic consequences. • Inadequate governance mechanisms exacerbate these issues. • Communities face a dilemma between staying in place and relocating.	Maldonado (2014)	• Adaptive governance structures are critical for supporting in-situ adaptation or community-led relocation efforts. • Multiple forms of knowledge need to be integrated into the adaptation process.
6.	Latin America	Local level	Analyse the impact of climate-induced relocation on Indigenous Peoples and assess the effectiveness of managed retreat as an adaptation strategy.	• Rising sea levels and increasing climate threats make climate-induced relocation necessary. • Poses significant risks, including the loss of cultural identity, community cohesion, and traditional knowledge	Felipe Pérez and Tomaselli (2021)	• Relocation processes indicate adverse impacts such as loss of community cohesion, cultural identity, and traditional knowledge among Indigenous Peoples. • Managed retreat faces resistance due to the potential loss of cultural heritage and social networks.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
7.	Chile	Local level	Examine climate change impacts and vulnerability trends among Mapuche communities from 1990 to 2015	• Trends in water scarcity, reduced agricultural production, vegetation colonisation, and population migration to higher altitudes due to climate change. • Increased pressure on Andean ecosystems and loss of traditional Mapuche knowledge	Parraguez-Vergara et al. (2016)	• Limited local capacity for adaptation • Governmental initiatives not fully meeting community needs • Critical need for local resilience-building efforts.
8.	Norway, Finland, Sweden, Russia	Regional level	Develop a framework for understanding the holistic effects of climate change on the Saami people.	• Significantly impacting Saami reindeer culture and traditional practices • The health of the Saami is slightly better than other populations in the region, but mental health risks related to changing environments are emerging concerns.	Jaakkola et al. (2018)	• Identified the interconnectedness of climate change impacts on cultural and health aspects among the Saami.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
9.	Bukavu, DR Congo	Local level	Explore vulnerability and adaptation strategies to climate change in local communities.	• The perception of climate change depends on observable impacts. • Climate-related risks compound existing vulnerabilities, worsening poverty. • Various adaptation measures are practised or planned, including local knowledge integration. • Implementation of adaptation faces significant barriers, such as financial constraints and lack of institutional support.	Bele et al. (2014)	• Participatory action research provided insights into local perceptions and realities of climate impacts, emphasising the need for context-specific adaptation strategies.
10.	South Africa	Local level	Investigate climate vulnerability and risks to an Indigenous Nama community.	• Climate hazards identified include drought, hot temperatures, and strong winds. • Lack of access to critical natural resources in the adjacent national park increases vulnerability.	Samuels et al. (2022)	• A remarkable disparity in access to natural resources impacts vulnerability levels. • The community reveals moderate vulnerability due to mixed adaptive capacity and sensitivity levels.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
11.	Australia	National/Regional	Assess climate change health impacts on Indigenous communities	• Extreme weather damages infrastructure and disrupts health services • Inadequate housing exacerbates heat stress and disease transmission • Climate impacts compound historical injustices and disrupt cultural connections to the Country	Hall and Crosby (2022)	• Indigenous health vulnerability is inseparable from connection to Country. • Adaptation requires addressing basic infrastructure needs and cultural wellbeing.
12.	South Australia	Regional	Examine vulnerability in remote Indigenous communities	• Climate change intensifies existing challenges (fire, floods, invasive species) • Traditional ecological knowledge is essential for adaptation • Climate risks must be understood within a broader socio-ecological context	Bardsley and Wiseman (2012)	• Effective adaptation requires integrating climate responses with broader social development and valuing Indigenous natural resource management.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
13.	Northland, Aotearoa, New Zealand	Regional	Evaluate the integration of Indigenous knowledge in climate planning	• Māori possess extensive ecological and disaster knowledge • Implementation challenged by systemic barriers and historical grievances • Common goals provide a pathway for knowledge integration	Mannakkara et al. (2023)	• Despite policy recognition of mātauranga Māori, practical implementation requires addressing colonial legacies and building trust-based partnerships
14.	Northern Aotearoa New Zealand	Community	Investigate diverse experiences of Māori women facing climate change	• Climate vulnerability varies significantly within Indigenous communities • Power dynamics affect adaptation capacity • Women's adaptation strategies include political self-determination and cultural renewal	Johnson et al. (2022)	• Indigenous climate vulnerability is heterogeneous; intersectional approach reveals how multiple identities influence adaptation capacity and strategies
15.	Taiwan	Local level	Understand the role of culture in the recovery process of relocating Indigenous communities through tourism livelihood.	• Appeal to culture, rather than land, as the foundation for community resilience and conflict reduction post-disaster. • Indigenous culture-based tourism can sustain livelihoods and support long-term development.	Lin and Lin (2020)	• The existing literature focuses primarily on rebuilding and renovating physical infrastructure. • There is a gap regarding the social aspects of recovery, particularly in indigenous perspectives.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
16.	Taiwan	Local level	Develop a livelihood vulnerability analytical framework and apply it to Indigenous communities regularly exposed to typhoons and geological hazards.	• Least vulnerable communities with abundant livelihood capitals • Intermediate vulnerability due to exposure to market and typhoon-related stresses • Most vulnerable communities are trapped in poverty and vulnerability cycles • A spatially explicit livelihood vulnerability index was developed to diagnose vulnerability dimensions and distribution.	Lin and Polsky (2016)	• Historical development strongly influences current vulnerability patterns. • Need for continuous longitudinal tracking of vulnerability dynamics for future research.
17.	Doi Mae Salong, Thailand	Mountainous landscape	Explore farmers' perceptions of climate change's impact on farming and adaptation measures.	• Changes in climate patterns affect farming negatively. • Adapt alternate farming practices using traditional techniques. • Higher vulnerability was observed at higher elevations compared to lower elevations.	Shrestha et al. (2017)	• Significant autonomous adaptation efforts are evident, yet challenges remain due to limited knowledge and financial resources.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
18.	Vietnam	National level	Study the livelihood vulnerability under climate change in rural areas.	• Higher LVI-IPCC value due to increased exposure to disasters like floods, sea storms, and tropical depressions, significantly impacting livelihoods.	Vo and Tran (2022)	• Emphasise the critical role of geographic location and exposure to climate-related hazards in determining vulnerability. • Highlight the importance of comprehensive datasets like VARHS in understanding multi-dimensional aspects of rural livelihood vulnerability.
19.	Sherpur and Mymensingh, Bangladesh	District level	Examine the impact of agricultural modernisation on sustainable livelihoods among tribal and non-tribal farmers.	• Tribal farmers generally had lower education levels than non-tribal farmers, affecting their adoption of modern agricultural practices. • Expressed the need for training in new technologies • Usage of modern agricultural practices was limited • Education, irrigation facilities, and community memberships positively influenced livelihood diversification.	Jannat et al. (2021)	• Limited educational opportunities among tribal farmers hindered the adoption of modern agricultural practices.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
20.	Nepal	District level	Analyse the micro-level vulnerability of the rural Chepang community	• Highlighting local-level differences in exposure and adaptive capacity. • Poor households with low adaptive capacity are consistently vulnerable regardless of location.	Piya et al. (2012)	• Importance of local-level vulnerability assessment for targeted policy measures
21.	Himachal Pradesh, India	Subdivision level	Develop a Tribal Household Livelihood Vulnerability Index focusing on Lahaul, Udaipur, and Spiti subdivisions in the western Himalayas cold deserts.	• The vulnerability of tribal households varies significantly across different socioeconomic and ecological conditions due to differences in adaptability, sensitivity, and exposure to climate change.	Kumar et al. (2023)	• The literature review showed significant variations in vulnerability metrics across Lahaul, Udaipur, and Spiti, influenced by socioeconomic and ecological factors unique to each subdivision.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
22.	Uttarakhand, India	Regional Level	Assess the trend of climatic variations, impacts, and adaptation strategies by the community.	• Periodic climatic fluctuations, including a downward annual and seasonal rainfall trend • Community perception notes reduced monsoon intensity, decreased winter snowfall, and erratic precipitation patterns. • Reduced crop yields and horticultural production due to climate change impacts	Pratap and Pratap (2021)	• Community adaptation strategies include varying crops, changing planting schedules, and enhancing water management practices.
23.	Gujarat, India	State level	Examine the livelihood conditions of tribal communities in Gujarat amidst rapid economic growth and widening disparities. Also, explore policy options for mitigating poverty among these communities, focusing on forest-based livelihoods and climate change adaptation.	• Significant poverty and deprivation among tribal communities despite economic growth • Dependence on forest resources for livelihoods, exacerbated by degradation and inadequate entitlements • Policy emphasis on forest regeneration, conservation, and sustainable management is crucial for poverty alleviation and climate resilience.	Shah and Sajitha (2009)	• Recognition of the Important role of forest resources in tribal livelihoods and the degradation challenges • Identification of policy gaps and the need for stronger entitlements and conservation incentives

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
24.	Madhya Pradesh, India	Local level	Evaluate climate change vulnerability and identify key drivers for effective adaptation strategies.	• Economic conditions, education level, and livelihood strategy significantly influence vulnerability more than social class. • Government programs must be inclusive and focus on the welfare of all social classes.	Yadava and Sinha (2020)	• The vulnerability index revealed critical insights into vulnerability drivers. • Household economic status and livelihood choices emerged as pivotal factors.
25.	Madhya Pradesh, India	Local level	Assess the impact of MGNREGA on vulnerability to climate change among tribal communities.	• MGNREGA has significantly reduced vulnerability related to climate variability, agriculture, water, and household economic conditions. • Increased water availability, diversified agriculture, improved crop yields and employment opportunities, and enhanced the economic status of tribal peoples.	Jha et al. (2017)	• The MGNREGA program effectively integrates natural and human systems, enhancing resilience to climate impacts.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
26.	West Bengal, India	Local level	Measure and compare livelihood vulnerability among Santal, Munda, Bhumij, and Lodha tribal communities in response to climate change.	• Tribal households exhibit greater sensitivity to climate change and have less capacity for adaptation. • The absence of alternative employment opportunities and planned activities exacerbates the impact of climatic variability on livelihoods, particularly agriculture-based ones.	Das and Basu (2022)	• It highlights the significance of beta regression modelling in determining vulnerability factors.
27.	West Bengal, India	Local level	Determine livelihood vulnerability among tribal farmers in the face of water scarcity.	• Income level, irrigation facilities, credit patterns, and social group participation strongly influence the higher LVI. • Monocrop cultivation and limited livelihood options augment vulnerability.	Mukherjee et al. (2022)	• A composite Livelihood Vulnerability Index model and IPCC-LVI technique were employed.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
28.	North-East India	Local level	Characterise Indigenous community-based climate vulnerability and capacity assessment in the Garo hills	• The Garo indigenous communities are highly vulnerable to climate change due to factors such as high poverty levels, dependence on natural resources, and frequent climatic hazards like cyclonic storms and droughts. • Climate change disrupts agriculture, water availability, and livelihoods.	Yadav and Sarma (2013)	• The study highlights strategic and short-term adaptation strategies rather than long-term planning.
29.	Nagaland, India	Local level	Explore the community-level vulnerability due to climate extremes and variability	• Minor exposure, sensitivity, or adaptive capacity changes could make the communities vulnerable. • Reducing sensitivity through safe housing infrastructure, food security, and sanitation development	Kuotsu et al. (2017)	• The Ao communities show somewhat higher vulnerability than the Angami tribes, but both are statistically vulnerable at similar levels.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
30.	Tripura, India	District level	Assess the extent of climate vulnerability using the LVI-IPCC technique among tribal and non-tribal populations in Tripura.	• Significantly impacts agricultural-based livelihoods. • Tribal households are more exposed to climate change impacts, leading to higher vulnerability than non-tribal households. • Exhibit greater sensitivity and lower adaptation capacity.	Roy et al. (2023)	• The LVI-IPCC method provides a structured approach to assessing vulnerability across demographic groups.
31.	Jharkhand, India	Village level	Estimate household vulnerability indices among tribal rural households	• Need for urgent income enhancement and livelihood improvement interventions. • Declining rainfall exacerbated vulnerability. • Efforts to improve irrigation and farm incomes showed limited impact.	Vatta et al. (2017)	• Declining rainfall exacerbated vulnerability, shifting additional households into more vulnerable categories. • Efforts to increase irrigation and farm incomes showed localised improvements but were insufficient, given the scale of vulnerability.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
32.	Jharkhand, India	Village level	Investigate the impact of climate change on NTFPs	• Increased maximum temperature has led to a significant decrease in NTFP yield. • Degraded the quality of NTFPs • Lack of incorporating Indigenous knowledge of climate change into institutional processes	Magry et al. (2023)	• Integrating Indigenous knowledge into adaptation strategies, addressing infrastructure development impacts, and enhancing institutional support for NTFP-based livelihoods
33.	Jharkhand, India	State level	Assess the impact of climate change on tribal economy and livelihoods	• Increased incidence of drought, affecting rain-fed agriculture and natural resource availability for tribal communities • Result in livelihood insecurity, decreased crop production, and increased vulnerability to diseases among crops and livestock	Barla (2016)	• Evidence shows a visible increase in drought frequency and intensity in Jharkhand, impacting tribal communities heavily dependent on natural resources and rain-fed agriculture.

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
34.	Jharkhand and Odisha, India	Village-level	Examine the perception of the Sauria Paharia community regarding climate change impacts on agroforestry and dietary diversity and assess adaptation strategies.	• Local climatic variability reduces crop productivity and food availability from forests and water bodies. • Declining agroforestry produce and diversity reduces household income, food insecurity, and migratory wage labour. • Use of climate-resilient Indigenous crop varieties, seed conservation, and reliance on Indigenous Forest foods during lean periods	Ghosh-Jerath et al. (2021)	• Limited resources and coping mechanisms of the Sauria Paharia community underline severe vulnerability to climate impacts. • Importance of promoting sustainable adaptation strategies to enhance resilience, income, and food security

Table 3 (continued)

Sl No.	Location	Spatial granularity	Objective	Main Findings	References	Literature Synthesis Insights
35.	Jharkhand, India	District level	Assess livelihood vulnerability among 15 tribes, including 8 Particularly Vulnerable Tribal Groups (PVTGs)	• Landless families are highly vulnerable. • Forest dwellers face challenges in education and healthcare access. • Minor forest produces and social welfare measures serve as crucial safety nets. • Improved access to banking, housing, and sanitation facilities • Need to enhance livelihood opportunities for forest-dwelling communities.	Behera et al. (2022)	• The study indicates the effectiveness of participatory approaches in assessing vulnerability. • Challenges in education and healthcare and the importance of safety nets are critical areas for policy intervention.

adaptation measures (Parraguez-Vergara et al. 2016). CC increases existing vulnerabilities, requiring integrated adaptation strategies focusing on essential sectors and decentralised governance structures to improve resilience and reduce the effect on economic growth and poverty reduction initiatives in the Democratic Republic of Congo, Africa (Bele et al. 2014). The Nama community in South Africa is moderately vulnerable to CC, mainly because of drought, high temperatures, and strong winds. It might increase as the harsh weather patterns become more severe. Therefore, it is important to use the existing co-management models with the South African National Park to mitigate the risks related to CC (Samuels et al. 2022).

In Australia, remote Indigenous communities face compounding climate vulnerabilities, with extreme weather events damaging essential infrastructure and disrupting health services. Hall and Crosby (2022) found that inadequate housing worsens heat stress and disease transmission, while climate impacts disturb cultural connections to Countries central to Indigenous well-being. In South Australia's Alinytjara Wilurara region, Bardsley and Wiseman (2012) observed that climate change intensifies existing challenges, including fire, floods, and invasive species management, illustrating that traditional ecological knowledge is necessary for effective adaptation strategies. In Aotearoa, New Zealand, Northland's local governments are working to integrate mātauranga Māori (Indigenous knowledge) into climate planning. Mannakkara et al. (2023) revealed that while Māori have extensive ecological and disaster knowledge, implementation is challenged by systemic barriers and historical grievances stemming from colonization. Johnson et al. (2022) demonstrated that climate vulnerability varies significantly even within Indigenous communities, with an intersectional approach showing how power dynamics affect adaptation capacity among Māori women, whose strategies often include political self-determination and cultural renewal as pathways to climate resilience.

In Asia, the communities least vulnerable to hostile effects are in the most isolated areas and have sufficient resources to reduce their vulnerability. The communities with moderate vulnerability face a dual risk from the market and natural hazards. On the other hand, the most vulnerable communities are trapped in a loop of poverty along with vulnerability (Lin and Polsky 2016). The Tsou Indigenous community in Taiwan experienced a resilient recovery after being relocated due to Typhoon Morakot due to their culturally based tourism livelihoods. It illustrates the importance of using indigenous culture, rather than land, as the foundation for community growth, as it promotes social resilience and decreases internal conflicts in the impact of a disaster (Lin and Lin 2020). The Doi Mae Salong region in Northern Thailand reveals farmers' awareness of CC's impact on agriculture and their strategies for coping, highlighting the economically poor population's vulnerability (Shrestha et al. 2017). The regions of North Central and South-Central Coasts in Vietnam are most vulnerable due to the increased frequency of floods, sea storms, and tropical depressions, which underlines the urgency for targeted interventions in these regions to protect rural livelihoods against climatic risks (Vo and Tran 2022). The study reveals tribal farmers in the Sherpur and Mymensingh districts of Bangladesh have lower education levels but are keen on technology training, indicating the potential for improved living standards (Jannat et al. 2021). The study in Nepal's rural Chepang community highlights the importance of targeted interventions in improving households' adaptive capacity, particularly among the poorest families, to mitigate climate variability (Piya et al. 2012).

The study highlights the need for sustainable livelihood policies in Himalayan tribal communities in India due to the varying vulnerabilities they face across socio-economic and ecological conditions (Kumar et al. 2023). The Jaunsar-Bawar tribal region in the Central Himalayas has been facing a decline in rainfall since the late 1980 s, necessitating institutional support for adaptation (Pratap and Pratap 2021). Rapid economic growth in Gujarat has worsened tribal community disparities, emphasizing the importance of sustainable resource management and forest-based livelihoods (Shah and Sajitha 2009). The study highlights the significance of socioeconomic factors like education, economic status, and livelihood type in assessing households' vulnerability to CC impacts in forest peripheral villages in Madhya Pradesh (Yadava and Sinha 2020). The Mahatma Gandhi National Rural Employment Guarantee Act 2005 (MGNREGA) programmes significantly decrease vulnerability to CC and poverty among tribal communities, enhancing income, employment opportunities, agricultural diversification, crop yield, and water availability (Jha et al. 2017). Policy interventions in West Bengal are suggested to address the vulnerability of tribal livelihoods to CC, considering factors such as agriculture income, family size, poverty, and water availability (Das and Basu 2022; Mukherjee et al. 2022). The Garo tribal communities in Meghalaya are at heightened risk due to high poverty, reliance on natural resources, and frequent disruptions to their livelihoods due to cyclonic storms and droughts (Yadav and Sarma 2013). Even minor exposure, sensitivity, or adaptive capacity changes could make communities vulnerable, demanding the enhancement of housing, food security, and sanitation (Kuotsu et al. 2017). Tribal households, closely dependent on agricultural activities, have greater sensitivity and lower adaptive capacity than non-tribal households, thus increasing their vulnerability to climate impacts (Roy et al. 2023). CC has significantly decreased the productivity and quality of non-timber forest products (NTFPs), showing the challenge faced by the lack of integrating Indigenous knowledge into institutional processes (Magry et al. 2023). The increasing incidence of drought and decline in forest cover in Jharkhand due to CC disproportionately affects the tribal communities, threatening their livelihoods and economy (Barla 2016). The Sauria Paharia tribal community is vulnerable to CC, reducing crop productivity, agroforestry diversity, and household food insecurity (Ghosh-Jerath et al. 2021). Therefore, executing sustainable adaptation strategies is crucial to improving resilience and ensuring food security for the community. Landless forest dwellers, including vulnerable tribal groups, in Jharkhand and Odisha face vulnerability due to inadequate access to education, healthcare, and social welfare measures (Behera et al. 2022). The necessity of focused development initiatives is highlighted by the need for comprehensive interventions to enhance income and livelihoods for tribal rural communities, with CC exacerbating levels of vulnerability (Vatta et al. 2017). These empirical analyses can identify best practices, lessons learned, and opportunities for scaling up adaptation efforts to enhance the resilience of tribal communities in the face of CC.

5 Adaptation and resilience strategies

In response to CC, tribal communities worldwide employ various adaptation and resilience strategies to cope with these changes and build sustainable futures. Various adaptation and resilience strategies are employed by tribal communities, focusing on indigenous knowledge, community-based approaches, and policy interventions. For example, Canada's First Nations communities have developed the concept of Two-Eyed Seeing, which promotes for integrating indigenous and Western knowledge systems to foster more inclusive and effective climate resilience strategies (Bartlett et al. 2012). Similarly, Ermine (2007) concept of Ethical Space provides a valuable foundation for helping respectful and equitable engagement between differing worldviews and governance systems.

5.1 Indigenous knowledge and practices

Indigenous people deeply understand their local environments and develop through generations of coexisting peacefully with nature (Carrin 2024). This conventional ecological knowledge forms the basis of many adaptation strategies that tribal communities utilise (Mallick et al. 2024; Parrotta and Agnoletti 2012). From agricultural practices to resource management techniques, indigenous knowledge systems offer constructive insights into sustainable living in changing climatic scenarios. Tribal communities mainly depend on agriculture and use traditional farming methods tailored to the local environment in response to CC. In arid regions, Indigenous communities are implementing water-saving methods such as rainwater harvesting, contour ploughing, and crop diversification to mitigate the effects of droughts and erratic rainfall patterns (Haddis 2018). Apart from agricultural practices, tribal communities have complex techniques for managing natural resources sustainably (Kala 2022). Many tribes have traditional forest management practices, including rotational grazing, selective logging, and forest fire management techniques (Charnley et al. 2007). These practices help maintain biodiversity, prevent soil erosion, and enhance ecosystem resilience in climate-related challenges (Lin and Lin 2020). Traditional knowledge holders pass down ancestral knowledge and rituals that raise social cohesion, spiritual well-being, and collective action (Xu et al. 2005). These cultural practices provide emotional support during harsh conditions and advance environmental stewardship and adaptive capacity within tribal communities.

5.2 Community-based adaptation approaches

Tribal communities use community-based adaptation (CBA) approaches to address CC impacts collaboratively (Fischer 2021). CBA highlight local participation, empowerment, and decision-making, enabling communities to recognize their vulnerabilities and develop context-specific adaptation strategies (McNamara and Buggy 2017). Tribal communities perform participatory assessments to recognize key climate risks and vulnerabilities (Long and Steel 2020; Mehta 2024). These assessments engage community members, local leaders, researchers, and non-governmental organizations (NGOs) to collect data, analyse findings, and prioritise adaptation actions (Salvador Costa et al. 2022; Samaddar et al. 2021). Integrating local and scientific knowledge in these assessments ensures that adaptation strategies are rooted in the realities of tribal livelihoods and environments.

Furthermore, many tribal communities are adopting ecosystem-based adaptation methods that harness the resilience of natural ecosystems to boost human well-being (Chaudhary et al. 2021; Panmei and Selvan 2024; Saikia et al. 2020). Ecosystem-based adaptation strategies include restoring degraded habitats, conserving biodiversity hotspots, and creating green infrastructure to buffer against climate impacts such as floods, storms, and heat waves (Inácio et al. 2020). By protecting and restoring natural ecosystems, tribal communities adapt to CC and contribute to global mitigation efforts (Chong 2014). However, CBA methods emphasise building social capital and encouraging community resilience through capacity building, knowledge exchange, and collective action (Carmen et al. 2022). Workshops, training programs, and community events were organised to raise awareness about CC, build adaptive skills, and strengthen social networks among tribal communities (Mallick et al. 2024; Mehta 2024; Nursey-Bray et al. 2022). CBA empowers community members to take charge of their adaptation processes, promoting self-reliance and long-term resilience (Mehta 2024; Samaddar et al. 2021).

5.3 Policy interventions

Tribal communities, despite their cultural heritage and ecological knowledge, frequently face challenges in adaptation due to limited resource access, land tenure insecurity, and insufficient government funding. Effective policy interventions are needed to address these challenges and build an assisting environment for tribal adaptation and resilience, as presented in Table 4. Indigenous rights to land, resources, and self-determination are crucial for tribal adaptation efforts, and governments must respect traditional governance systems and involve tribal communities in decision-making processes (Brugnach et al. 2017). In addition, policymakers should develop inclusive and participatory policy frameworks that incorporate indigenous knowledge systems, cultural values, and community priorities into

Table 4 Policy interventions for enhancing tribal resilience

Sl. No.	Policy Intervention	Description	Advantages	References
1.	Integration of Indigenous Knowledge Systems	Incorporation of traditional knowledge into climate change adaptation and resilience strategies	• Enhances cultural significance and acceptance of adaptation strategies • Influences time-tested, locally adapted practices • Strengthens community engagement and ownership	Mehta (2024)
2.	Community-Based Natural Resource Management	Policies advancing community ownership and management of natural resources	• Promotes sustainable use of resources • Empowers communities • Improves resource conservation and resilience against climate impacts	Mishra et al. (2021)
3.	Capacity Building Initiatives	Training programs and workshops are needed to improve the capacity of tribal communities to adapt.	• Builds local skills and knowledge • Increases community resilience • Enhances ability to implement and maintain adaptation measures	Nursey-Bray et al. (2022)
4.	Access to Climate Information and Early Warning Systems	Establishment of systems for spreading climate information and early warning systems	• Provides timely information to prevent and mitigate climate impacts • Reduces loss of life and property • Enhances preparedness and response capabilities	Panda et al. (2023)
5.	Sustainable Livelihood Diversification	Policies encouraging diversification of livelihoods to decrease dependency on climate-sensitive sectors	• Reduces risk of income loss from climate events • Enhances economic stability • Promotes resilience by spreading risk across multiple sectors.	Mehta (2024)
6.	Land Tenure Security	Measures to secure land rights for tribal communities, preventing land catching and displacement	• Ensures long-term access to land and resources • Reduces conflicts over land • Provides a stable source for livelihood activities and investment	Naik and Madhanagopal (2022)
7.	Disaster Risk Reduction Strategies	Implementation of strategies to decrease vulnerability to climate-induced disasters	• Reduce damage and loss from disasters • Enhances community safety • Reduces economic costs associated with disaster recovery	Kumar et al. (2023)
8.	Financial Support Systems	Provision of financial support, grants, and subsidies for climate adaptation programs.	• Facilitates implementation of adaptation strategies • Reduces financial burden on communities • Encourages innovation and practical adaptation	Craig (2022)
9.	Institutional Support for Indigenous Governance Structures	Recognition and cooperation for Indigenous governance systems in decision-making processes	• Strengthens governance and representation • Ensures culturally applicable solutions • Enhances authenticity and compliance with adaptation measures	Lefthand-Begay et al. (2024)
10.	Research and Data Collection	Funding and support for research initiatives to collect data on climate change impacts on tribal livelihoods	• Informs evidence-based policymaking • Identifies specific vulnerabilities and needs • Supports targeted and effective adaptation strategies	Mishra et al. (2019)

national and regional adaptation plans (Wheeler and Root-Bernstein 2020). Governments, international organizations, and development agencies should support tribal communities in implementing adaptation projects and initiatives by mainstreaming Indigenous perceptions into climate policies (Mehta 2024). It should also include funding for climate-resilient infrastructure, livelihood diversification programs, disaster preparedness training, and climate information packages tailored to the needs of tribal communities.

6 Challenges and lacunae

Understanding the challenges, identifying the research gaps and practice, and interpreting the vulnerability of tribal communities to CC is necessary for developing effective adaptation and mitigation strategies. The challenges faced by the government, policymakers, and NGOs in implementing CC adaptation and mitigation strategies for tribal communities are detailed in Fig. 6.

6.1 Data and research limitations

The limited availability and accessibility of reliable data are one of the primary challenges in evaluating the vulnerability of tribal communities to CC. Many tribal regions lack robust monitoring methods and data collection infrastructure, indicating gaps in understanding CC's specific impacts on these communities. Furthermore, there is a need for more interdisciplinary research that incorporates traditional ecological knowledge with scientific approaches. CC impacts vary spatially and temporally, posing challenges in correctly capturing and predicting these dynamics at the regional level (Nissan et al. 2019). Most existing studies focus on broader regional or national scales, neglecting the nuanced vulnerabilities of specific tribal groups residing in diverse ecological sites. Several authors faced various challenges regarding data availability such as difficulty in data collection due to the remote, tribal-dominated areas, including language barriers and limited village access (Magry et al.

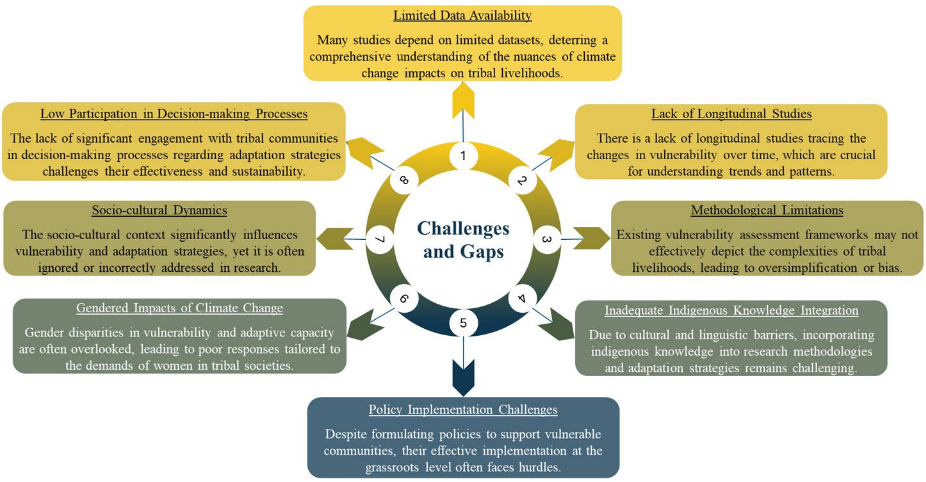


Fig. 6 Summary of Challenges and Gaps in Existing Research

2023); reliance on peer-reviewed journal articles, excluding other relevant literature related to tribal communities (Fan et al. 2022); unavailability of data at a consistent scale for all required variables; inconsistencies in data availability, with some variables being spatially continuous and others only available at administrative levels (Kasthala et al. 2024); lack of significant quantifiable literature on indigenous agricultural practices in India; and absence of direct studies assessing CC risk (Aich et al. 2022).

6.2 Policy and implementation barriers

Tribal communities often encounter marginalisation in decision-making practices and policy formulation related to CC adaptation and mitigation (Ahmed et al. 2022; Fischer 2021; Mallick et al. 2024). The lack of indigenous representation leads to policies that may inadequately direct the exceptional needs and priorities of tribal communities. Moreover, limited financial resources for CC adaptation and mitigation strategies in tribal regions augment vulnerabilities (Hazarika et al. 2024; Mallick et al. 2024; Mehta 2024; Mishra et al. 2019). Insufficient funding limits the performance of adaptation strategies, such as infrastructure development, capacity building, and community-based initiatives (Fischer 2021). However, many CC adaptation policies and programs follow top-down methods that do not consider the knowledge, preferences, and priorities of local tribal communities. Various studies highlighted policy and implementation barriers in CC adaptation, emphasizing the need for culturally sensitive, context-specific approaches, effective stakeholder engagement, tangible benefits, capacity-building, and integration of Indigenous knowledge to confirm the resilience and empowerment of rural and Indigenous communities (Datta and Kairy 2024; Nkoana et al. 2018; Salvador Costa et al. 2022; Samaddar et al. 2021).

6.3 Socio-cultural challenges

Rapid modernisation and globalisation augment cultural erosion, leading to a loss of identity and social cohesion among tribal communities (Ozer and Obaidi 2022). Additionally, gender disparities within tribal communities' composite vulnerability to CC effects. Women often show the impact of environmental degradation and resource scarcity. However, in decision-making processes about adaptation and attempts to create resilience, their views are typically marginalised (Matsa 2021). Several studies have demonstrated significant socio-cultural challenges in their research on tribal communities, including struggles with cultural identity, traditional sickness perceptions affecting healthcare access, the integration of indigenous knowledge in sustainability efforts, and barriers to education and economic development (Mallick et al. 2024; Maske et al. 2015; Mishra 2023; Yoganandham 2023).

Overcoming data limitations, advancing inclusive policy frameworks, and promoting socio-cultural resilience are critical steps towards confirming the well-being and sustainability of tribal communities in a changing climate. Addressing the challenges mentioned above is crucial for effectively developing the resilience of tribal communities to CC.

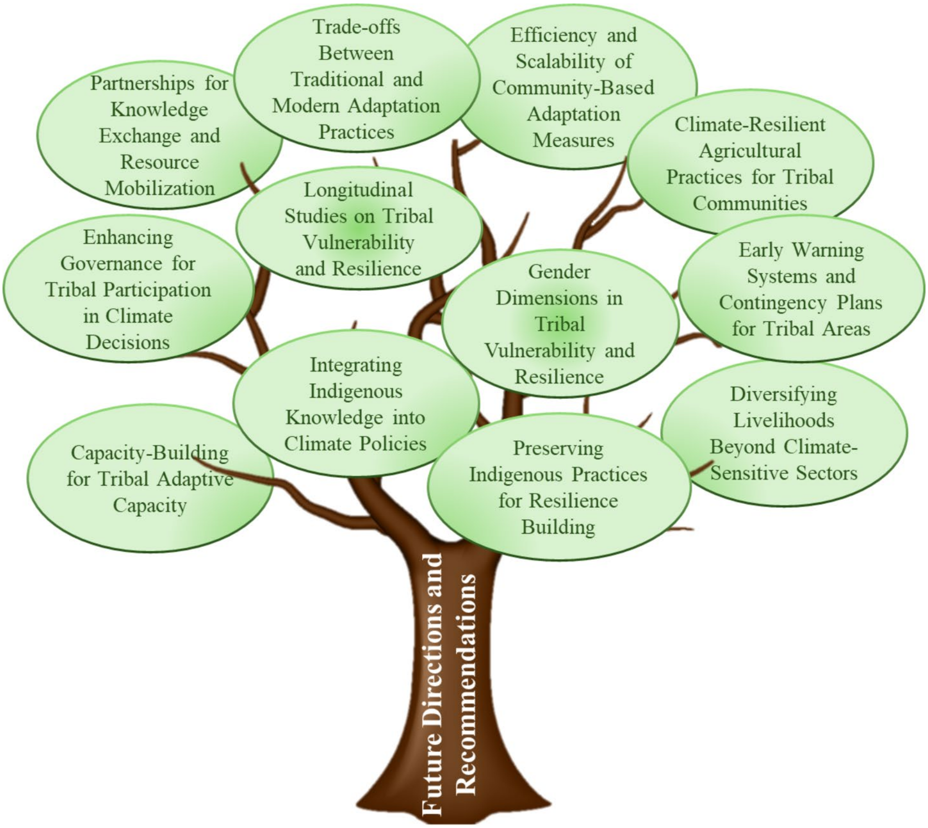


Fig. 7 Recommendations for Future Research and Policy Development

7 Future directions and recommendations

Understanding the connection between CC effects and tribal communities is necessary for developing future approaches to enhance resilience and reduce vulnerability. Figure 7 depicts recommendations for future research and policy development. Furthermore, an essential research priority is to improve data collection methodologies developed for the specific circumstances of tribal communities. It includes combining traditional knowledge systems with modern scientific techniques to augment the accuracy and relevance of data. Ray (2023) highlights the ability to integrate Traditional Knowledge Systems (TKS) with modern scientific approaches to tackle livelihood security, conserving biodiversity, and sustainable development goals. Strengthening the paradigm of Science and Technology (S&T) and promoting resilience and interconnection can be achieved by incorporating indigenous concepts with multidisciplinary techniques. In the same way, Rahmah and Sulistyono (2024) indicate the necessity of integrating indigenous climate mitigation measures within institutional CC policies. Tribal communities in Indonesia have used their extensive traditional knowledge systems and local wisdom to create innovative techniques to cope with the vulnerabilities caused by CC. Longitudinal studies following CC impacts and livelihood strategies over time are essential for understanding the dynamic nature of vulnerability. Collaborative efforts between researchers, community members, and local authorities are critical for developing widespread datasets that capture the multifaceted dimensions of livelihood vulnerability. Researchers can recognise emerging challenges and assess the effectiveness of adaptation interventions by recording trends and patterns. These studies should prioritise community engagement and participatory approaches to confirm that the voices of tribal communities are listened to and respected throughout the research process. There is a need for multi-disciplinary research methods that combine understandings from diverse fields such as climatology, ecology, anthropology, economics, and sociology.

Challenges faced by government, policymakers, and NGOs while implementing strategies reveal that recognising and protecting Indigenous rights, including land tenure, resource management, and cultural heritage preservation, should be included in the policy. Enhancing indigenous governance structures empowers communities to emphasise their sovereignty and participate actively in decision-making processes associated with climate adaptation and natural resource management (Muchunguzi 2023). Governments and development agencies should prioritise integrating indigenous knowledge systems into CC adaptation planning and implementation (Machingura et al. 2018). Tribal communities possess insightful knowledge about local ecological systems and adaptive practices that can complement scientific knowledge and enhance the effectiveness of adaptation initiatives (Ahmed et al. 2022; Mallick et al. 2024; Mishra et al. 2019). Efforts should be made to enable knowledge exchange and collaboration between traditional practitioners and outside stakeholders (Mehta 2024). Investments in community-based adaptation measures are crucial for building resilience at the local level (Chaudhary et al. 2021; Mishra et al. 2019; Panda et al. 2023). Governments and contributors should allocate resources to support community-driven projects that enrich livelihood diversification, strengthen natural resource management systems, and improve access to climate-resilient infrastructure and technologies (Akchurina 2022). Capacity-building programs should prioritise empowering vulnerable groups, including women, youth, and marginalised communities (Sethi et al. 2017). Integrating CC factors into broader development policies and programs is essential for mainstreaming adaptation strategies. Development interferences should be planned to strengthen the ability to adapt and reduce vulnerability across all sectors, including agricultural systems, water management, health, and infrastructure development (Salas and Yepes 2020). Overall, it appears that challenges associated with CC need a collaborative effort of stakeholders, researchers, policymakers, practitioners, and community members.

8 Conclusions

This study has shed light on the relationship between CC and the vulnerability of tribal communities' livelihoods. A review of available literature showed the impacts of CC on several dimensions of tribal livelihoods, including agriculture, water resources, forests, health, and socioeconomics. It also revealed the different adaptation and resilience strategies used by tribal communities, leveraging indigenous knowledge, community-based approaches, and policy interventions. However, it is evident that challenges continue, including data limitations, policy gaps, and socio-cultural barriers. There is an urgent need for collaborative efforts among researchers, policymakers, and tribal communities to address these challenges and improve the adaptive capacity of tribal livelihoods in the face of CC. It is advised that governments and development organizations give tribal communities' rights top priority and safeguard them while also incorporating indigenous knowledge systems into CC adaptation planning and implementation. CC can contribute to developing more resilient and sustainable futures for tribal communities.

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Declarations

Ethical approval All authors have read, understood, and have complied as applicable with the statement on"Ethical responsibilities of Authors" as found in the Instructions for Authors.

Competing interest We have nothing to declare.

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